Machine Learning A (2026)
Home Assignment 2

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1 Preprocessing

1.1 Importance of Preprocessing

When measuring income in thousands of USD the nearest neighbor algorithm *would not* lent C money as person B is closest (note that any strictly increasing monotone transformation preserves ordering):

$$\|(A-C)\|^2 = (47-21)^2 + (35-36)^2 = 677,$$

$$\|(B-C)\|^2 = (22-21)^2 + (40-36)^2 = 17.$$

If we instead measure income in dollars the characteristics of person A is now closer to C and BoL *would* lent the money to person C:

$$\|(A-C)\|^2 = (47-21)^2 + (35000-36000)^2 = 1,000,676,$$

$$\|(B-C)\|^2 = (22-21)^2 + (40000-36000)^2 = 16,000,001.$$

The conclusion is that choices doing preprocessing generally effects an algorithms predictions.

1.2 Input Centering

1.2.1 (a)

Applying matrix product rules yields:

$$\begin{aligned}
\gamma \boldsymbol{X} &= \left(\boldsymbol{I} - \frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)\boldsymbol{X} \\
&= \boldsymbol{X} - \frac{1}{N}\mathbf{1}\mathbf{1}^\top\boldsymbol{X} \\
&= \boldsymbol{X} - \mathbf{1}\left(\frac{1}{N}\boldsymbol{X}^\top\mathbf{1}\right)^\top \\
&= \boldsymbol{X} - \mathbf{1}\bar{\boldsymbol{x}}^\top \\
&= \boldsymbol{Z}
\end{aligned}$$

The last equality is seen by looking at an arbitrary row i of $\gamma\boldsymbol{X} =$

$\boldsymbol{X} - \mathbf{1}\bar{\boldsymbol{x}}^\top$:

$$\begin{bmatrix} x_{i1} - \frac{1}{N}\sum_{n=1}^{N} x_{n1} & \cdots & x_{id} - \frac{1}{N}\sum_{n=1}^{N} x_{nd} \end{bmatrix} = \left[\boldsymbol{x}_i - \bar{\boldsymbol{x}}\right]^\top = \boldsymbol{z}_i^\top,$$

where x_{ij} denotes the element of $\gamma\boldsymbol{X} = \boldsymbol{X} - \mathbf{1}\bar{\boldsymbol{x}}^\top$ at row i and column j. That is

$$\gamma\boldsymbol{X} = \boldsymbol{X} - \mathbf{1}\bar{\boldsymbol{x}}^\top = \begin{bmatrix} \text{——} & \mathbf{z}_1^\top & \text{——} \\ \text{——} & \mathbf{z}_2^\top & \text{——} \\ & \vdots & \\ \text{——} & \mathbf{z}_N^\top & \text{——} \end{bmatrix},$$

which is the very definition of $\boldsymbol{Z}$.

1.2.2 (b)

For every column c in $\boldsymbol{Z}$ the last coordinate (or any one coordinate i just choose the last for notational convenience) equals the negative sum over the $N-1$ other elements of the column c:

$$\begin{aligned}
\sum_{n=1}^{N-1} z_{nc} &= x_{1c} - \frac{1}{N}\sum_{n=1}^{N} x_{nc} + x_{2c} - \frac{1}{N}\sum_{n=1}^{N} x_{nc} + \cdots + x_{(N-1)c} - \frac{1}{N}\sum_{n=1}^{N} x_{nc} \\
&= \sum_{n=1}^{N-1} x_{nc} - \frac{1}{N}\left((N-1)\sum_{n=1}^{N} x_{nc}\right) \\
&= \sum_{n=1}^{N-1} x_{nc} - \sum_{n=1}^{N} x_{nc} + \frac{1}{N}\sum_{n=1}^{N} x_{nc} \\
&= -\left(x_{Nc} - \frac{1}{N}\sum_{n=1}^{N} x_{nc}\right) \\
&= -z_{Nc}.
\end{aligned}$$

Thus the whole last row is a linear function of the $N-1$ rows

$$\boldsymbol{z}_N^\top = -\sum_{n=1}^{N-1} \boldsymbol{z}_n^\top,$$

and such the $r(\boldsymbol{Z}) < N$, where we have implicitly used the equivalence of row rank and column rank.

1.3 Input Whitening

1.3.1 (a)

Calculating we get $Var(x_1) = Var(\hat{x_1}) = 1$ and

$$\begin{aligned} Var(x_2) &= Var\left(\sqrt{1-\epsilon^2}\hat{x_1} + \epsilon\hat{x_2}\right) \\ &= (1-\epsilon^2)Var(\hat{x_1}) + \epsilon^2 Var(\hat{x_2}) + 2\left(\sqrt{1-\epsilon^2}\epsilon\right) Cov\left(\hat{x}_1, \hat{x}_2\right) \\ &= 1 - \epsilon^2 + \epsilon^2 \\ &= 1, \end{aligned}$$

where we have used the independece of $\hat{x}_1$ and $\hat{x}_2$ implying $Cov\left(\hat{x}_1, \hat{x}_2\right) = 0$. Finally

$$\begin{aligned} Cov(x_1, x_2) &= Cov\left(\hat{x}_1, \sqrt{1-\epsilon^2}\hat{x_1} + \epsilon\hat{x_2}\right) \\ &= Cov\left(\hat{x}_1, \sqrt{1-\epsilon^2}\hat{x_1}\right) + Cov\left(\hat{x}_1, \epsilon\hat{x}_2\right) \\ &= \sqrt{1-\epsilon^2} Var\left(\hat{x}_1\right) + \epsilon Cov\left(\hat{x}_1, \hat{x}_2\right) \\ &= \sqrt{1-\epsilon^2}, \end{aligned}$$

where we again used the independece of $\hat{x}_1$ and $\hat{x}_2$.

1.3.2 (b)

Using that $x_2 = \sqrt{1-\epsilon^2}\hat{x_1} + \epsilon\hat{x_2} \iff \hat{x_2} = \frac{x_2}{\epsilon} - \frac{\sqrt{1-\epsilon^2}}{\epsilon}\hat{x}_1$ we obtain (for $\epsilon \neq 0$)

$$\begin{aligned} f(\hat{\boldsymbol{x}}) &= \hat{w}_1\hat{x}_1 + \hat{w}_2\hat{x}_2 \\ &= \hat{w}_1 x_1 + \hat{w}_2\left(\frac{x_2}{\epsilon} - \frac{\sqrt{1-\epsilon^2}}{\epsilon}\hat{x}_1\right) \\ &= \underbrace{\left(\hat{w}_1 - \hat{w}_2\frac{\sqrt{1-\epsilon^2}}{\epsilon}\right)}_{w_1} x_1 + \underbrace{\frac{\hat{w}_2}{\epsilon}}_{w_2} x_2. \end{aligned}$$

1.3.3 (c)

Applying the formula derived above setting $\hat{w}_1 = \hat{w}_2 = 1$ yields

$$f(\hat{\boldsymbol{x}}) = 1 \cdot \hat{x}_1 + 1 \cdot \hat{x}_2 = \left(1 - \frac{\sqrt{1-\epsilon^2}}{\epsilon}\right) x_1 + \frac{1}{\epsilon} x_2.$$

Therefore this target function is available whenever $C \geq \left(1 - \frac{\sqrt{1-\epsilon^2}}{\epsilon}\right)^2 + \left(\frac{1}{\epsilon}\right)^2$ implying that the maximum amount of regularization is obtained at

$$C_{min} = \left(1 - \frac{\sqrt{1-\epsilon^2}}{\epsilon}\right)^2 + \left(\frac{1}{\epsilon}\right)^2, \quad \epsilon \neq 0.$$

1.3.4 (d)

Both terms in C_{min} goes to infinity as $\epsilon \to 0$ so

$$C_{min} \to \infty \text{ as } \epsilon \to 0.$$

2 Hoeffding's Bound

Firstly we repeat Hoeffding's inequality for convenience.

$$\mathbb{P}\left(\sum_{i=1}^{N} X_i - \mathbb{E}\left[\sum_{i=1}^{N} X_i\right] \geq \epsilon\right) \leq \exp\left\{-\frac{2\epsilon^2}{\sum_{i=1}^{N}(b_i - a_i)^2}\right\}. \tag{2.1}$$

Immediately we have $X \in [-2, 1]$, so we choose $a = -2$ and $b = 1$. ϵ is determined by the condition

$$S \geq 22 \iff S - \mathbb{E}[S] \geq 22 - \mathbb{E}[S] = \epsilon,$$

where $S = \sum_{i=1}^{100} X$. So we calculate the expected value of S:

$$\mathbb{E}[S] = \sum_{i=1}^{100} \mathbb{E}[X] = \frac{100}{3}(-2 + 0.8 + 1) = -\frac{100}{15}.$$

That is $\epsilon = 22 - \left(-\frac{100}{15}\right) = \frac{22 \cdot 15 + 100}{15}$. Inserting this into yields the probability bound

$$\mathbb{P}(S > 22) \leq \mathbb{P}(S \geq 22) \leq \exp\left\{-\frac{2 \cdot \left(\frac{22 \cdot 15 + 100}{15}\right)^2}{100 \cdot 3^2}\right\} \approx 0.161.$$

3 Illustration of Markov's, Chebyshev's, and Hoeffding's Inequalities

3.a

1. Using numpy i sample 1,000,000 repetitions of the stochastic variable $M_{20} = \frac{1}{20}\sum_{i=1}^{20} X_i$ and evaluate the empirical frequency given by the assignment for the different alphas. The results are plotted as the indigo line in figure 1.

2. The intervals of 0.05 is sufficient because M_{20} only take on these values as it is a mean of 20 Bernoulli distributed stochastic variables.

3. Applying Markow's inequality yields

$$\mathbb{P}(M_{20} \geq \alpha) \leq \frac{\mathbb{E}[M_{20}]}{\alpha} = \frac{0.35}{\alpha}.$$

The bound for the α's is plotted with an orange line in figure 1

4. The condition $M_{20} \geq \alpha$ does not permit the direct application of Chebyshew's inequality. Therefore we manipulate the condition a bit (conserving the measure / probability of the event) and then apply the inequality

$$\begin{aligned}
\mathbb{P}\left(M_{20} \geq \alpha\right) &= \mathbb{P}\left(M_{20} - \mathbb{E}\left[M_{20}\right] \geq \alpha - \mathbb{E}\left[M_{20}\right]\right) \\
&\leq \mathbb{P}\left(\left|M_{20} - \mathbb{E}\left[M_{20}\right]\right| \geq \alpha - \mathbb{E}\left[M_{20}\right]\right) \\
&\leq \frac{\mathbb{V}\left(M_{20}\right)}{\left(\alpha - \mathbb{E}\left[M_{20}\right]\right)^2} \\
&= \frac{0.35(1-0.35)}{20(\alpha - 0.35)^2},
\end{aligned}$$

where the last equality used the i.i.d property and the well known formula for the variance of a Bernoulli distributed variable

$$\mathbb{V}\left(M_{20}\right) = \mathbb{V}\left(\frac{1}{20}\sum_{i=1}^{20} X_i\right) = \frac{1}{(20)^2}\sum_{i=1}^{20}\left(\mathbb{V}\left(X_i\right)\right) = \frac{1}{20}\mathbb{V}(X_1) = \frac{0.35(1-0.35)}{20}.$$

5. Like with Chebyshew's inequality we can not directly apply Hoeffding's inequality to the condition. Therefore we again manipulate the condition and then apply the inequality

$$\begin{aligned}
\mathbb{P}\left(M_{20} \geq \alpha\right) &= \mathbb{P}\left(M_{20} - \mathbb{E}\left[M_{20}\right] \geq \alpha - \mathbb{E}\left[M_{20}\right]\right) \\
&= \mathbb{P}\left(\sum_{i=1}^{20} X_i - \mathbb{E}\left[\sum_{i=1}^{20} X_i\right] \geq 20\left(\alpha - \mathbb{E}\left[M_{20}\right]\right)\right) \\
&\leq \exp\left\{-\frac{2\left(20\left(\alpha - \mathbb{E}\left[M_{20}\right]\right)\right)^2}{\sum_{i=1}^{20}(1-0)^2}\right\} \\
&= \exp\left\{-40\left(\alpha - \mathbb{E}\left[M_{20}\right]\right)^2\right\}.
\end{aligned}$$

The resulting bounds are again graphed in figure 1 now with a purple line.

6. Comparing the graphs in figure 1 we firstly note that the calculated frequency based on simulated data stays below the bounds.

We also clearly observe the superiority of Chebyshew's and Hoeffding's inequality to Markow's inequality in bounding the probability of the event $M_{20} \geq \alpha$. In addition we observe how the bound Hoeffding's inequality provides shrinks faster (exponentially) compared to the bound found by applying Chebyshew's inequality.

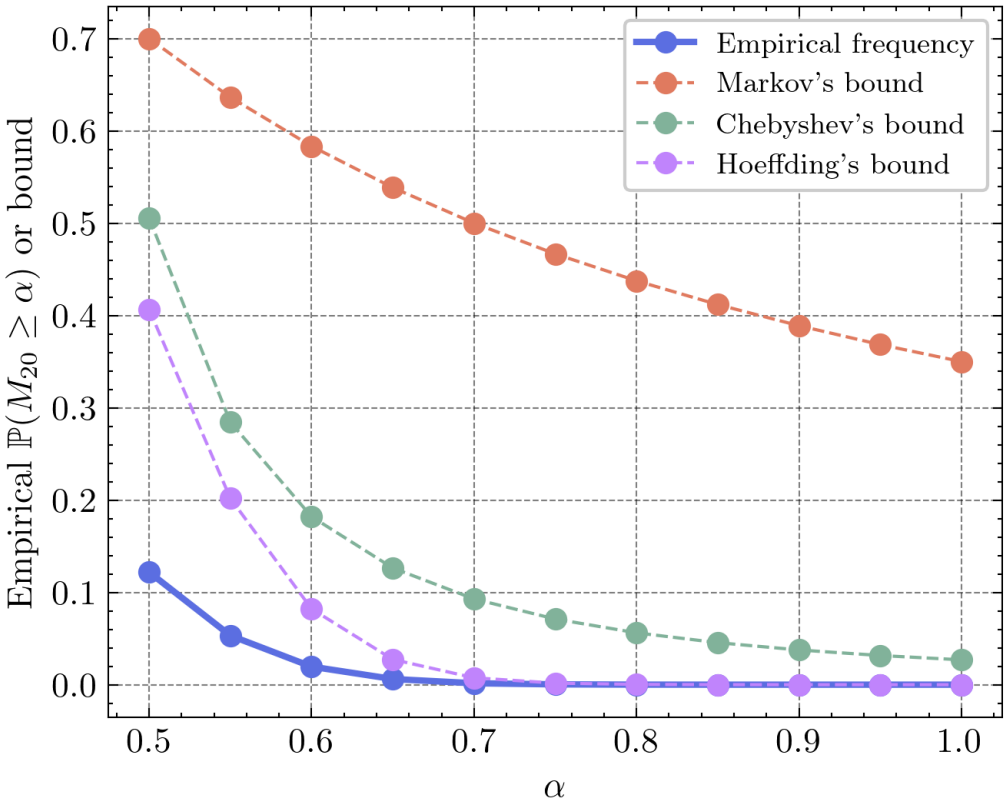


Figure 1: Frequency and Different bound on event

Note: The dotted lines represent probability bounds on the event $\mathbb{P}(M_{20} \geq \alpha)$ and the indigo filled in line graphs the empirical frequency of the event for $N = 1,000,000$ experiments.

7. For $\alpha = 1$ we have

$$\mathbb{P}(M_{20} \geq 1) = \mathbb{P}\left(\sum_{i=1}^{20} X_i = 20\right) = 0.35^{20} \approx 7.61 \cdot 10^{-10}.$$

using the binomial coefficient $\binom{n}{k} = \frac{n!}{(n-k)!k!}$ we calculate the probability for $\alpha = 0.95$

$$\begin{aligned}
\mathbb{P}(M_{20} \geq 0.95) &= \mathbb{P}\left(\sum_{i=1}^{20} X_i \geq 19\right) \\
&= \mathbb{P}\left(\sum_{i=1}^{20} X_i = 19\right) + \mathbb{P}\left(\sum_{i=1}^{20} X_i = 20\right) \\
&= \binom{20}{19}(0.35)^{19}(0.65) + 0.35^{20} \\
&\approx 2.90 \cdot 10^{-8}.
\end{aligned}$$